

Lecture 3 Team Projects

Duration, Convexity, and Portfolio Strategies

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1 Project 1: Immunize a Known Liability

Scenario. Fund a known payment in five to eight years with two or three government bonds.

Objectives. Match asset and liability present value and duration; explain local protection; stress-test parallel and nonparallel curve moves.

Sources. U.S. Treasury, TreasuryDirect, Bank of Canada, or an instructor set of prices and cash flows.

Deliverables. Liability assumptions, bond weights, PV/duration/convexity calculations, three shocks, rebalancing note, and a 10–12 minute presentation.

Rubric (100 points). Framing 15; calculations 30; shocks 25; interpretation 20; presentation and auditability 10.

2 Project 2: When Convexity Earns Its Keep

Scenario. Decide when paying for higher convexity is worthwhile for two duration-matched alternatives.

Objectives. Construct a matched pair; compare duration-only, duration-plus-convexity, and full repricing; relate value to shock size, volatility, and carry.

Sources. U.S. Treasury, Bank of Canada, FINRA Fixed Income Data, or instructor option-free bond snapshots.

Deliverables. Matched comparison, five-shock repricing, approximation-error chart, benefit-versus-cost conclusion, and a 10–12 minute presentation.

Rubric (100 points). Matching 20; analytics 25; repricing 25; interpretation 20; presentation and sources 10.